Project Design Phase-II
Technology Stack (Architecture & Stack)

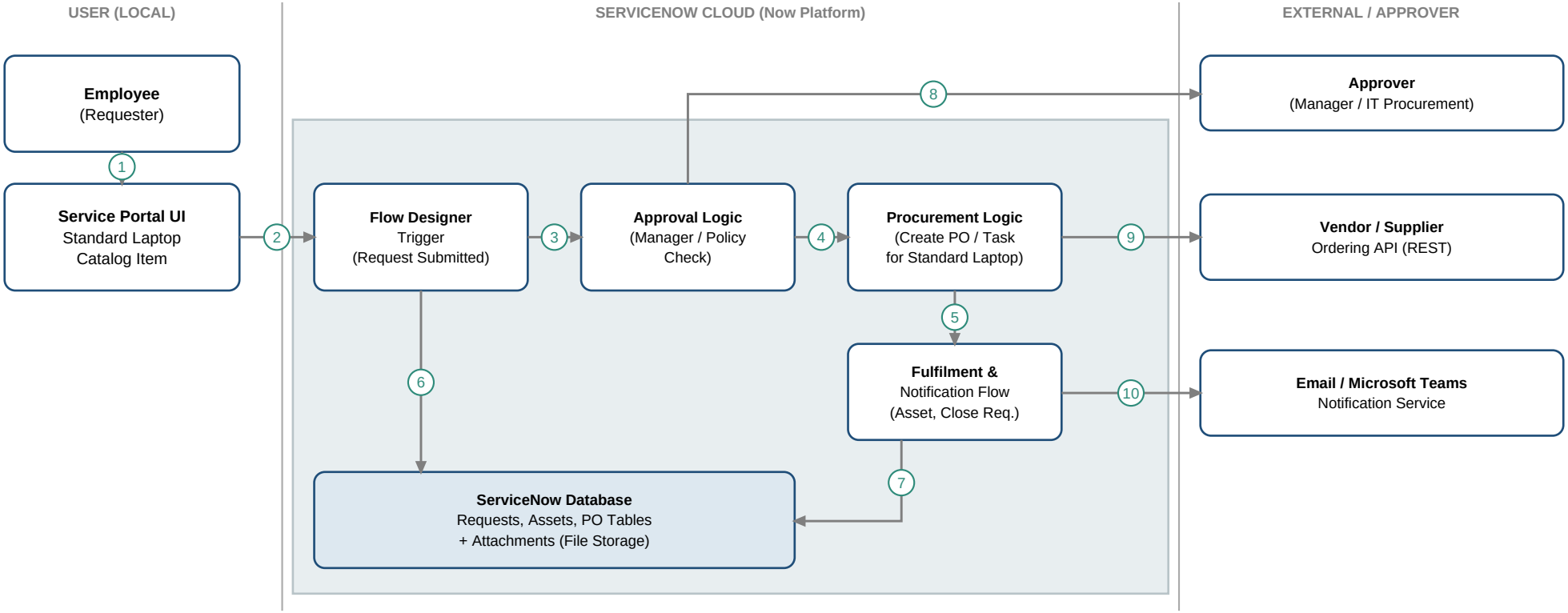
Date	29 September 2026
Team ID	SWTID-2026-5078
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Automating standard laptop orders with Flow Designer

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Flow: 1 Employee opens the catalog 2 Request submitted to Flow Designer 3 Standard-model / policy check 4 Approval outcome triggers procurement 5 Order handed to fulfilment 6-7 Records, assets and attachments stored in the database 8 Approval request sent to approver 9 Purchase order sent to vendor API 10 Status notification to requester

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employees request a standard laptop through the Service Portal catalog item (Web UI on desktop and mobile browsers).	HTML, CSS, JavaScript, ServiceNow Service Portal (AngularJS widgets)
2.	Application Logic-1	Flow Designer flow triggered when the standard laptop catalog request is submitted.	ServiceNow Flow Designer (Trigger, Actions)
3.	Application Logic-2	Approval logic: checks the standard model / policy and routes the request for approval.	Flow Designer Approval action, JavaScript (Business Rules / Script)
4.	Application Logic-3	Procurement and fulfilment logic: creates the purchase order and tasks, updates the asset and closes the request.	Flow Designer Subflows & Actions
5.	Database	Request, catalog item, asset and purchase order records, configurations.	ServiceNow platform tables (relational database)
6.	Cloud Database	Database service on cloud hosted by the ServiceNow instance.	ServiceNow Now Platform cloud database
7.	File Storage	File storage requirements: quotes, invoices and approval attachments.	ServiceNow Attachments (platform file storage)
8.	External API-1	Sends the purchase order to the vendor / supplier and receives order status.	REST API via IntegrationHub / REST Message
9.	External API-2	Sends status notifications to the requester and approvers.	Email / Microsoft Teams notification integration
10.	Machine Learning Model	Not applicable in the current scope of the project.	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Local Server Configuration: Not applicable (no local server required) Cloud Server Configuration: ServiceNow cloud instance (Now Platform)	Cloud (ServiceNow instance)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	AngularJS, Bootstrap, JavaScript (used by ServiceNow Service Portal)
2.	Security Implementations	Role-based access controls (ACLs) on requests and approvals, encrypted communication and authenticated API calls.	Role-based ACLs, HTTPS / TLS encryption, OAuth / Basic Auth for REST, Audit logs
3.	Scalable Architecture	Multi-tenant cloud platform with reusable flows, subflows and actions that can be extended to other catalog items.	Now Platform (cloud), Modular Flow Designer subflows
4.	Availability	Cloud-hosted instance with redundant infrastructure and failover managed by the platform.	ServiceNow cloud data centers (redundancy & failover)
5.	Performance	Automated, asynchronous flow execution reduces manual effort and turnaround time; indexed tables for fast lookups.	Flow Designer async execution, Table indexing

References:

https://c4model.com/
https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
https://www.ibm.com/cloud/architecture
https://aws.amazon.com/architecture
https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d